Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: Wt 32800259

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a =$ 27.03 $\quad b =$ 1.67 $\quad r =$ .67

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

   (a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$P(t) = 12.75(1.041)^t$

   (b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$P(25) = 12.75(1.04)^{25} = \$33.9894$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

   (a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

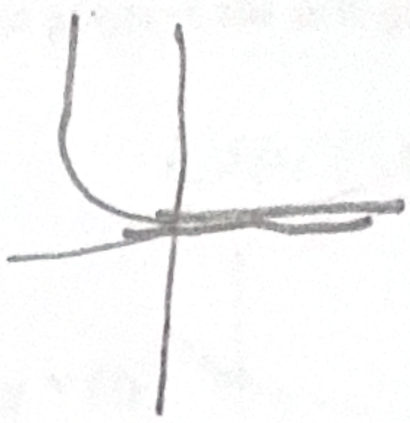
$P(t) = 6500 - 60(t)$

   (b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500(.95)^t$

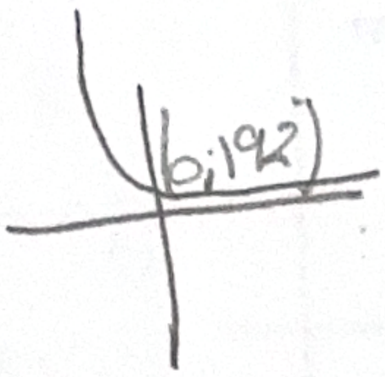
4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

$y = ab^t \quad t \to \infty$

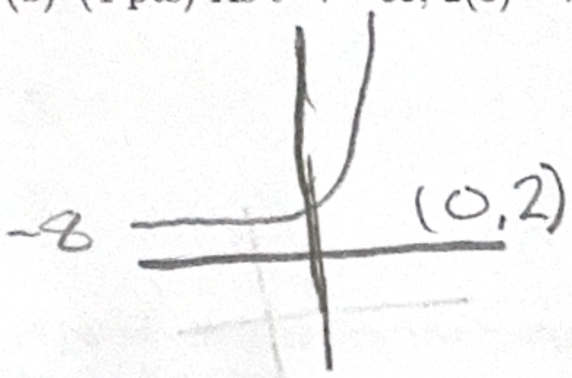


5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ $\emptyset$

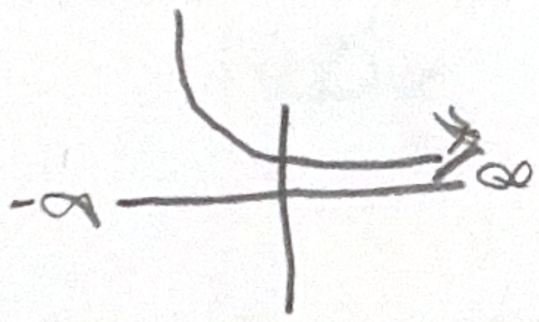


(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ $\emptyset$

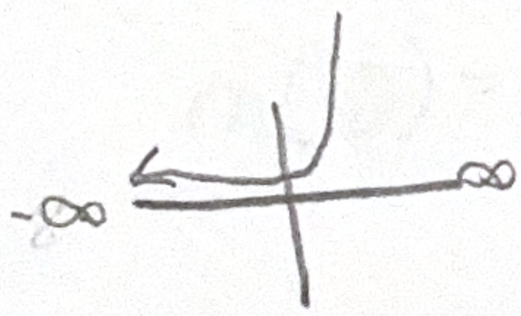


(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$

$f(t) = 5(3+(.09)^t$
$= 5(3+(.09+1)^t$
$5(3+1.09$

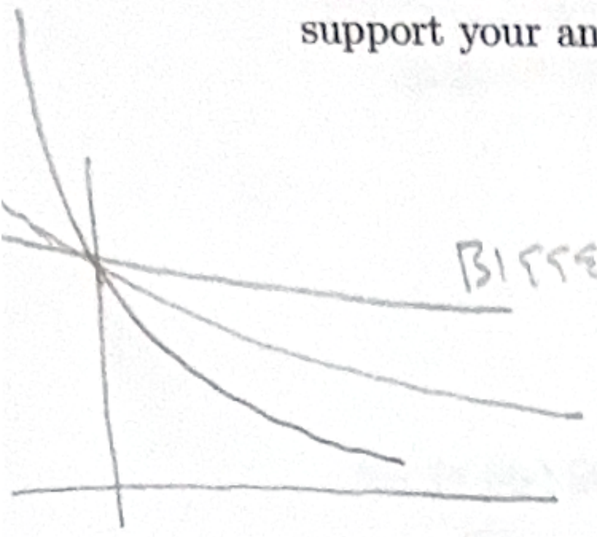


(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ $\emptyset$



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1+r)^t$ and $g(t) = b(1+s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.

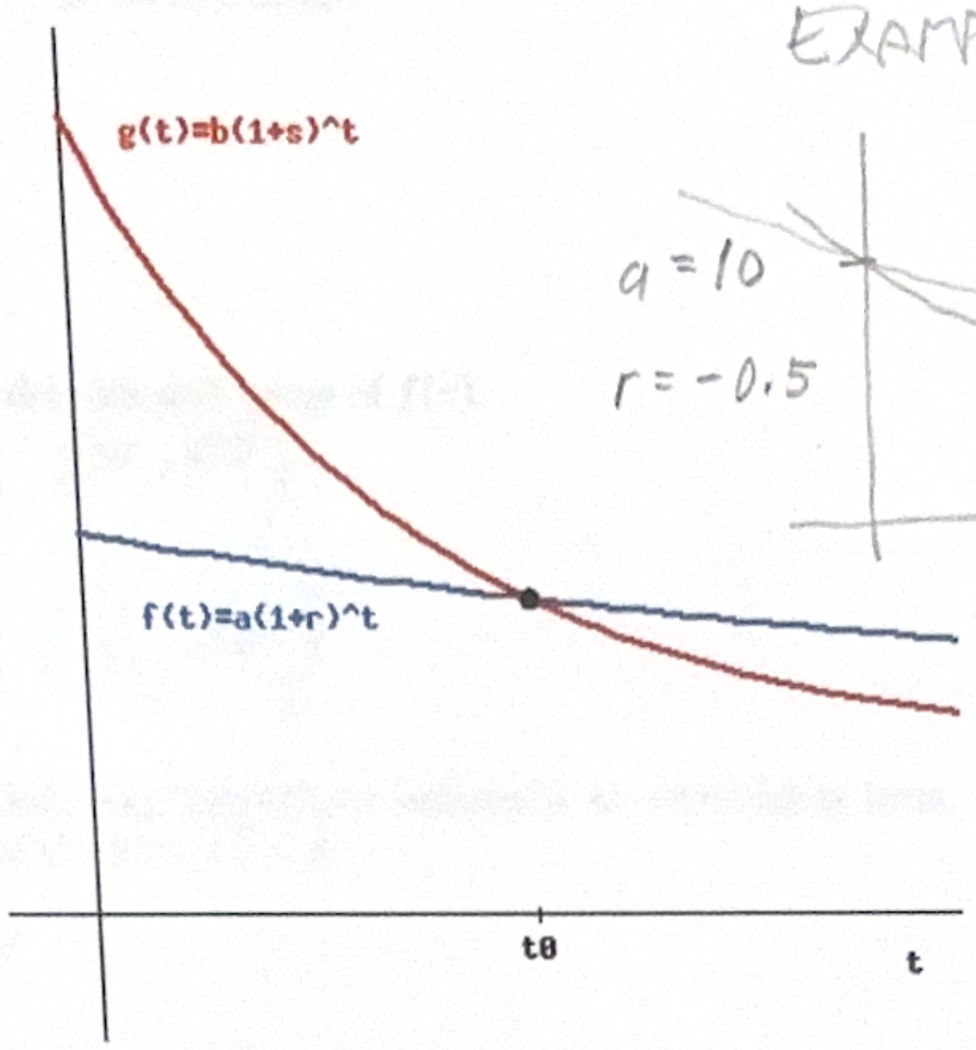
NEAR r

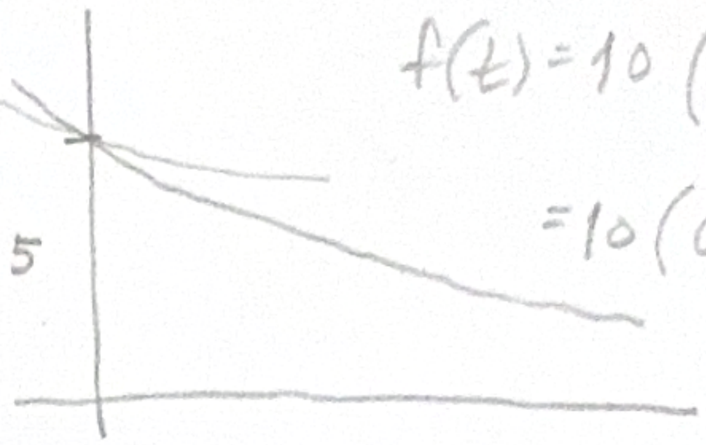


$f(t) = a(1+r)^t$

SINCE THE GRAPH IS DECREASING, $1+r<1$, AND THUS r IS NEGATIVE. IF r DECREASES (BECOMES MORE NEGATIVE) $1+$

EXAMPLE





$a = 10$

$r = -0.5$

$f(t) = 10(1+(-0.5))^t$

$= 10(0.5)^t$

IF r INCREASES TO -0.2:

$f(t) = 10(1+(-0.2))^t$

$= 10(0.8)^t$

$f(t) = ab^t$

7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

160g

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025 \cdot 10} = \frac{160}{e^{.25}} = \boxed{124.6081}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

   (a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

   (b) (4 pts) State the domain and range of $f(x)$.

Domain: $(-\infty, \infty)$

Range: $(2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

   (a) (4 pts) $\ln 10 = y$

   (b) (4 pts) $\ln x = 5$

$x = e^5$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

   (a) (4 pts) $\ln(e^{1.7})$

1.7

   (b) (4 pts) $e^{\ln 4}$

4

   (c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\ln(e^{-3})$

-3

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$-8 \quad -8$

$-\ln(2-x) = -7$

1

$\ln(2-x) = 7$

$e^{\ln(2-x)} = e^7$

$2 - x = e^7$

$-2 \quad -2$

$-x = e^7 - 2$

1

$x = -e^7 + 2$

(c) (4 pts) $4^x = 3$

$\ln(4)^x = .$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

2

$4^{2x-4} = 30$

$\ln(4)^{2x-4} = \ln(30)$

$2x-4 \ln(4) = \ln(30)$

$\ln(4)$

$2x - 4 = \frac{\ln(30)}{\ln(4)}$

$+4 \quad +4$

$2x = \frac{\ln(30)}{\ln(4)} + 4$

$2 \quad 2$

$x = 3.2267$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.